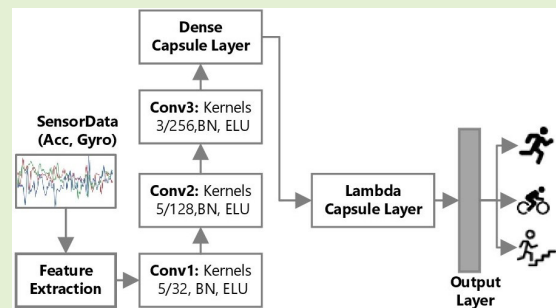


Sensors-Based Human Activity Recognition Using Hybrid Features and Deep Capsule Network

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Abstract—With the continuous advancements in artificial intelligence, human activity recognition (HAR) technologies have garnered widespread attention and found applications across diverse domains. Recently, various features and deep learning (DL) models are proposed for HAR using sensors data. Though the existing models and features have achieved notable performance, their recognition accuracy needs to be enhanced and computational cost needs to be reduced. This research introduces a novel integration of various features and deep capsule network (DeepCapsNet) capable of real-time signal processing. To construct the HAR model, the information about patterns shape and symmetry of the sensor data distribution, and variations in frequency modulation was extracted. This information was given to the DL model (DeepCapsNet) that integrates several convolutional layers (CLs) and DeepCapsNet. The CLs in DeepCapsNet are used to process temporal sequences and deliver scalar outputs, while the capsule network is utilized to retrieve the equivariance which enhances the performance of HAR model. At last, the efficiency of the DeepCapsNet is comprehensively assessed in comparison to other baseline models using three benchmark HAR datasets. The average accuracy of DeepCapsNet for UCI HAR, WISDM, and PAMAP2 datasets are 97.6%, 98.5%, and 99.9%, respectively. This research findings revealed the effectiveness of the DeepCapsNet over the performance of the baseline models in terms of accuracy and computational cost. The feature selection and model optimization need to be further explored to enhance the performance of HAR model.

Index Terms—Deep capsule network (DeepCapsNet), human activity recognition (HAR), Mel frequency cepstral coefficients (MFCCs), multifeatures, wearable sensors.



I. INTRODUCTION

HUMAN activity recognition (HAR) is an automatic process to recognize and classify specific behaviors and

movements from sensors data [1]. In recent years, HAR research using wearable sensors has gained attention from both industries and academia. The advancement of semiconductor technologies has facilitated the integration of inexpensive sensors with small size, lightweight, and low power consumption into wearable devices [2]. Presently, HAR system using unobtrusive sensing equipment has found extensive applications in sports [3], smart home environments [4], security [5], medical diagnosis [6], and robotics [7]. The significance of HAR system lies in its capability to deliver real-time monitoring and tracking of human activities, support and noteworthy feedback for numerous practical applications. For example, a HAR system can be deployed in healthcare to track the movements and actions of disabled/elderly persons and to come up necessary support. In the context of sports, HAR systems serve the purpose of monitoring and analyzing athlete's movements to detect areas for improvement. Similarly, in the field of robotics, the HAR systems can be deployed to facilitate human-robot interactions. The HAR process conventionally includes three primary stages: collection of data, extraction of salient features, and classification [8]. In the initial stage, data

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are gathered from different modalities, including accelerometers and gyroscopes. In the following step, the signal data provides the discriminative features including statistical, time-based, and frequency-based features. The final phase employs classification of algorithm deriving features into distinct activity classes, including running, walking, standing, and sitting.

The extraction of suitable and efficient features is an important requirement to recognize various human activities. Thus, machine learning (ML) algorithms heavily rely on heuristic manually designed features. In contrast, deep learning (DL) models can automatically derive high-level representations from low-level time-series data [9]. Recent studies about DL models for HAR have reported superior performance compared to traditional ML approaches [10]. In DL models, convolutional neural network (CNN) and long short-term memory (LSTM) are commonly employed. These both models are used to derive temporal and spatial features. Nonetheless, CNN could only obtain the invariance with pooling and scalar outputs instead of equivariance. Alternately, the deep capsule network (DeepCapsNet) is one of the algorithm utilized to derive spatial features. DeepCapsNet with vector and routing output can grasp the equivariance and showed the remarkable result on MNIST dataset [11], [12]. The DeepCapsNet employed in various areas such as text classification, license plate recognition [13], speech emotion recognition [14], and text classification. Moreover, the gated recurrent unit (GRU) [15], a variant of LSTM, offers a simpler structure with fast convergence rate and less parameters. Attention mechanism [16] combined with the GRU is an information selection mechanism and it is used to assign varied weights to the separate regions of GRU. The GRU model has been utilized in various applications such as question detection, health monitoring, and stock trend prediction [17], [18].

The main issue in HAR system is to address the complexity and variability of activities [19]. The activities exhibit substantial diversity with regards to context, intensity, and duration, posing challenges to construct accurate and vigilant HAR systems. Moreover, the selection of sensors and placement on the body significantly improve the effectiveness of HAR systems. Like, wearable sensors can yield precise data but could pose intrusion upon the user, whereas nonwearable sensors can offer user convenience at the cost of less accurate data. Currently, ever increasing interest in the development of HAR systems which may operate within real-world settings, characterized by unstructured and unpredictable activities. One strategy to tackle this issue is to utilize the integration of multimodal signals, where signals from several sensors are fused to extend a more comprehensive perspective for the ongoing activities [20]. The key contributions of this study, based on the above analysis are as follows.

- 1) This work integrates a variety of features to capture different aspects (spectral, statistical, and frequency-related information) of the activity signal. This integration enhanced the discriminative power of the model to distinguish between different activities more effectively.
- 2) This work utilized a model that used convolutional layers (CLs) and capsule networks to get local spatial information and to preserve the spatial relationships among information. This combination facilitated accurate recognition of extracted features with low computational cost.
- 3) The proposed features and prediction model is evaluated using the benchmark, complex, and freely available datasets (UCI-HAR, WISDM, and PAMP2).
- 4) A comparative evaluation is conducted between the performance of the DeepCapsNet model and the hybrid features against baseline techniques and features.

The study is organized as follows: Section II investigates the prior research about DL for HAR. Section III provides an in-depth exploration of the feature extraction process, the classification algorithm employed, and the evaluation metrics utilized. Section IV explains the outcomes derived from numerous experimental analyses conducted under various experimental conditions. At last, the Section V provides the conclusion of this study.

II. RELATED WORK

Extensive research efforts devoted for the recognition of physical activity comprising of the data acquired through smartphones sensors such as accelerometers and gyroscopes. The procedure of data collection involves utilizing smartphone sensors to capture human activities during routine tasks, including sitting, standing, and walking, along with cell phone placed in the human belongings. Presently, several research studies have focused on the sensor-based HAR. These studies employed either traditional ML algorithms or DL models to recognize human activities. A study [21] proposed a methodology where features were captured from gyroscope and accelerometer sensor data to predict the activities. The authors employed support vector machine (SVM) and k -nearest neighbor (k -NN) algorithms to predict the activities collected in two publicly available datasets.

In the domain of HAR, various DL models like CNN, LSTM, and their integration have been investigated. Notably, the variants of CNN have been reported to recognize activities by capturing information from inertial sensors data. For instance, [22] presented a deep CNN for HAR utilizing gyroscope and accelerometer sensors embedded in smartphones. This DL model exploited the temporal dependencies of 1-D time-series data and captured raw sensor data for robust features. In order to select suitable parameters for CNN, [23] proposed a 1-D CNN for HAR and conducted a comprehensive investigation for seven metaheuristic approaches. The study utilized the UCI-HAR dataset to assess the algorithm's optimization. The obtained results demonstrated the strength of metaheuristic algorithms in optimizing CNN parameters effectively. To capture the diverse array of receptive fields in feature layer, [24] introduced a CNN architecture employing the idea of hierarchical split (HS). Comprehensive studies were carried out using benchmark datasets, revealing that the HS approach delivered notable performance enhancements when compared to baseline models across several datasets, including PAMAP2, WISDM, UCI-HAR, and UNIMIB-SHAR. Moreover, study

exhibited extraction of discriminative features by utilizing multiscale receptive fields.

Recurrent neural networks (RNNs) are specifically designed to derive temporal dependencies in time series data. Thus, many variants of RNN family such as LSTM and BiLSTM have been proposed to address the challenges of HAR [25]. The study [26] introduced different types of deep RNNs for the recognition of human activities. The proposed method employed various architectures including both unidirectional and bidirectional deep RNNs for HAR tasks. Similarly, [27] proposed a bidirectional LSTM architecture for HAR utilizing time-series data obtained from a mobile positioned on the individual's waist. The best accuracy achieved by this model reaches 93.8%.

Despite the significant performance achieved by the above-mentioned DL models, their recognition accuracy needs further improvement. Thus, recent studies have combined CLs and LSTM to construct hybrid deep network to identify activities of human. For instance, the study [28] introduced a DeepConvLSTM comprised of CLs and LSTM layer. The model is constructed to automatically extract features and model the temporal dependencies from multimodal wearable sensors. Thakur et al. [29], investigated a DL approach for HAR with mobile accelerometer and gyroscope data. Three DL models namely autoencoders, CNN and LSTM were combined in this approach. CNN was employed to derive the discriminative information, while autoencoders were utilized to degrade the dimensions of the data and temporal modeling was performed using the LSTM model. The ConvAE-LSTM was assessed on WISDM, UCI-HAR, OPPORTUNITY, and PAMAP2 datasets. The findings of the demonstrated approach enhanced the recognition output with regard to computational time and accuracy over baseline models. Another research [30] explored the environmental contextual factors, specifically illumination and noise levels, to support sensor data utilizing a hybrid CNN-LSTM method. The reported method involved the fusion of comprehensive background information along with the foundational low-level sensor data to improve the accuracy of recognition. The research conducted two distinct experiments: the initial experiment involved the utilization of triaxial signals to train the baseline models, on the other hand, the second experiment incorporated both contextual data and triaxial sensing signals. The results exhibited that the inclusion of contextual information resulted in significantly improved accuracy in comparison to the baseline HAR models.

To address the limitations of CNN, Geoffrey Hinton introduced a capsule network, and it demonstrates superior performance over baseline models on the MNIST dataset [31]. Subsequently, capsule network has been utilized in various domains, including speech emotion recognition [32]. The latest studies on HAR have employed capsule networks. For example, [33] proposed a capsules network for identification of the spatial relationship in order to control the deficiencies of prevalent models. Another study [34] introduced a capsules network called 1-D-HARCapsNet for classifying human activities. The authors evaluated the proposed network on the WISDM database. Though the model has a simple architecture like the conventional capsule network and also achieved high

TABLE I
SUMMARY OF LITERATURE ON HAR

Study	Approach	Key Findings
[21]	ML algorithms using gyroscope and accelerometer data	Employed SVM and k-NN for activity prediction from smart-phone sensor data.
[22]	Deep CNN utilized gyroscope and accelerometer sensors	The authors demonstrated deep CNN's ability to capture temporal dependencies in inertial sensor data to improve the HAR model accuracy.
[23]	Metaheuristic optimization for 1D CNN	Seven metaheuristic approaches were investigated to optimize 1D CNN parameters to achieve the improved performance on the UCI-HAR dataset.
[24]	CNN architecture with hierarchical split (HS)	A CNN architecture using hierarchical split (HS) was introduced to capture diverse receptive fields to enhance the performance across benchmark datasets.
[25]	LSTM, BiLSTM	Various deep RNN architectures, LSTM and BiLSTM were explored for HAR and achieved 93.8% accuracy.
[28]	Hybrid Deep-ConvLSTM	DeepConvLSTM model was introduced that combined CLs and LSTM to extract features and temporal dependencies from multi-modal wearable sensor data.
[29]	CNN, autoencoders, and LSTM	Combined autoencoders, CNN, and LSTM to derive discriminative information, reduce data dimensions, and model temporal dependencies.
[30]	Hybrid CNN-LSTM	Contextual information was fused with sensor data using a CNN-LSTM hybrid model. The model achieved significantly improved recognition accuracy compared to baseline models, particularly when environmental contextual factors were considered.

accuracy, but it was tested for one dataset only. Notably, the surveyed models incorporate shallow capsules network, suggesting potential for further improvement in recognition performance. The summary of literature review is presented in Table I.

III. PROPOSED METHODOLOGY

A typical process of HAR involves three key stages: the activity-specific features from signal data; the construction of a model for activity recognition from these signals; and the testing phase to evaluate a predicted activity matches the expected pattern. Subsequently, extensive research has been performed in all three areas. A feature set was generated

TABLE II
STATISTICS OF THE UCI HAR DATASET

Activity	Data	Percent
SITTING	1777	17.25%
LYING	1944	18.87%
STANDING	1906	18.50%
DOWNSTAIRS	1406	13.65%
UPSTAIRS	1544	14.99%
WALKING	1722	16.72%
TOTAL	10299	100%



TABLE III
STATISTICS OF THE WISDM DATASET

Activity	Data	Percent
UPSTAIRS	122869	11.19%
WALKING	424399	38.64%
DOWNSTAIRS	100427	9.14%
JOGGING	342179	31.16%
STANDING	48395	4.41%
SITTING	59939	5.46%
TOTAL	1098208	100%



by deriving salient information from the collected activity signals. The HAR model was constructed utilizing this feature set as input to a DeepCapsNet. Using various evaluation matrices, the performance of the proposed HAR system was assessed. Moreover, the model's effectiveness was examined by testing it against a distinct set of test data and baseline HAR methods. A detailed explanation of these strategies is provided in Sections III-A–III-E, as illustrated in Fig. 1.

A. Dataset Description

The suggested model was assessed using three benchmark datasets: UCI-HAR, PAMP2, and WISDM. These three datasets comprise motion data obtained from sensors, covering a wide range of activities. The UCI-HAR is a prime dataset designed for evaluating the performance of newly developed HAR models. It includes six activities, namely SITTING, LYING, STANDING, WALKING DOWNSTAIRS, WALKING UPSTAIRS, and WALKING. Each activity sample in this dataset comprised of nine channels. Table II provides a summary of the UCI-HAR.

The second benchmark dataset is wireless sensor data mining (WISDM), is a widely recognized dataset containing 1 098 208 acceleration measurement data points which are classified with heterogeneous distributions into six activities as given in Table III. The signals of all activities are shown in Fig. 2. Moreover, a heatmap in Fig. 3 reveals the correlation between activities signals, where each column of WISDM dataset is illustrated to reveal the anomalies initiated by any particular column. The statistical analysis shown in Fig. 3 reveals that the x -axis, y -axis, and z -axis columns are similar.

The final dataset employed in this article is the PAMP2, which comprises data collected from sensors placed on people's hands, chests, and ankles. This dataset integrates data from accelerometers, magnetometers, gyroscopes, temperature, and other wearable sensors [35]. It includes nine subjects participating in 18 activities, with six of them being optional. The experiments recognized only 12 of these activities. The

TABLE IV
STATISTICS OF PAMAP2 DATASET

Activity	Data	Percent
STANDING	17350	10.01%
WALKING	21786	12.57%
LYING	17600	10.15%
IRONING	21770	12.56%
SITTING	16929	9.77%
CYCLING	15044	8.68%
NORDIC_WALK	17198	9.92%
CLEANING	16028	9.25%
DOWNSTAIRS	9593	5.53%
UPSTAIRS	10709	6.18%
ROPE_JUMPING	451	0.26%
RUNNING	8887	5.13%
TOTAL	173345	100%



data points in each activity in the PAMAP2 are listed in Table IV. No substantial correlation between other data and gyroscope sources due to the statistical similarities among various columns in the PAMAP2 dataset. Consequently, the data from gyroscope sensor is omitted from further analysis. In addition, correlations are observed between accelerometer and temperature data, as well as between heart rate and chest magnetometer data.

B. Data Preprocessing

The collected raw sensors data are segmented into sequences through a particular window size. These segmented sequences encapsulate essential information about various activities and are employed as input to extract the features. The segmentation methods include both fixed and adaptive window sizes. This study adopted a fixed window size of 128 for segmentation and overlapping ratio of 50% for WISDM dataset to create a set of segments to extract the spectral and temporal features, i.e., $S = \{S_1, S_2, \dots, S_N\}$, where N represents the number of segments. Each segment is characterized as a sequence of particular length (K) containing data collected from wearable devices and sensors, such as triaxial accelerometers, as defined in the following equation:

$$\text{Segment} = [D_i, D_{i+1}, \dots, D_{i+K-1}] \quad (1)$$

where, D_i denotes the data in the i th period.

C. Feature Extraction

The extraction of meaningful and distinct information from signals plays an important role in accurately recognizing human activities. Recently, experts have recommended various characteristics for HAR. Among these, short-time characteristics such as perceptual linear predictive (PLP) and Mel frequency cepstral coefficients (MFCCs) have gained significant attention due to their ability to capture repeated patterns and signal effectiveness. The effectiveness of MFCCs features in recognizing human activities has been reported in numerous experiments. However, in the case of complex collections of activity signals, these features tend to degrade, making it challenging to capture specific activity features. This research introduces a novel approach that combines temporal and

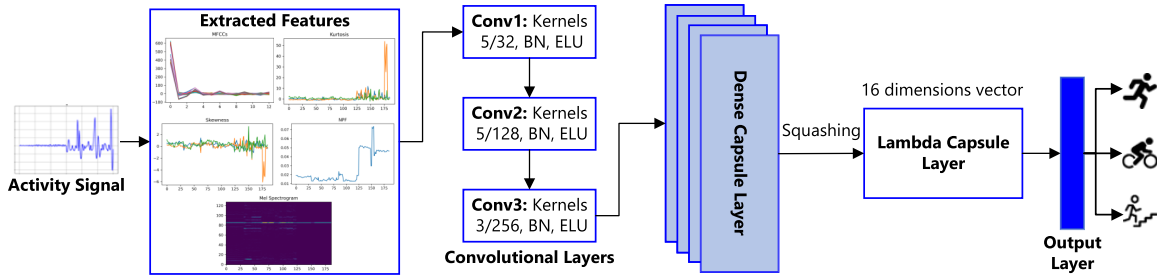


Fig. 1. Proposed Methodology for HAR of.

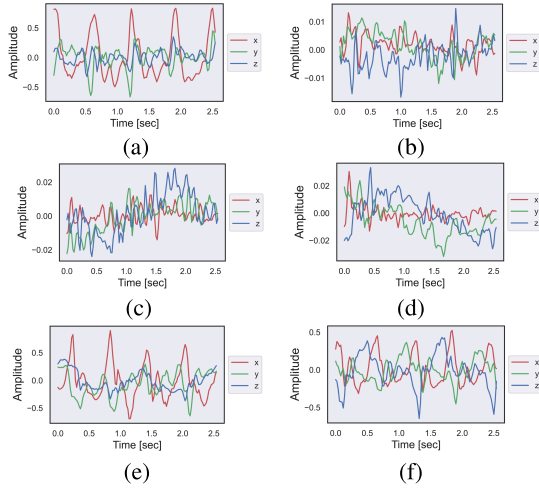


Fig. 2. Activities signals from WISDM dataset. (a) Downstairs. (b) Lying. (c) Sitting. (d) Standing. (e) Upstairs. (f) Walking.

spectral features to enhance the accuracy of HAR techniques. Temporal features encompass the extraction of statistical metrics from activity signals, offering valuable insights into the temporal evolution of the signal. These features offer benefits such as minimal computational demands and suitability for real-time human activities recognition. Conversely, spectral features identify the energy distribution within the activity signal. Therefore, it is essential to consider window sizes of d_t seconds, denoted by $N = f_s \times d_t$ samples, for extracting diverse feature vectors. In this case, f_s depicts the sampling frequency. The fusion of extracted features is described below.

MFCCs are used in the HAR field [36] and signal processing. To derive MFCCs feature, the activity signal is first divided into fixed length frames. Second, a windowing method is executed to lower the silence from all frames of activity signal. To capture important signal characteristics across lower and higher resonance frequency, digital filters come in both linear and logarithmic variants

$$\text{Mel}_{\text{freq}}(f) = 2595 \times \log_{10} \left(1 + \frac{f}{700} \right). \quad (2)$$

The spectral frequency (f) of the modified signal is transformed into a logarithmic scale referred to as the “Mel,” which is influenced by pitch of activity signal. The “Mel” is utilized to determine all values computed from the transformed activity signal using Fast Fourier Transform. Subsequently, the power logs are computed at each Mel-frequency and are converted back to time series by employing Discrete Cosine Transform.

At last, obtained amplitudes from resultant spectrum are called MFCCs.

Mel-spectrogram is an important feature as it captures relevant frequency patterns inherent in human activities. The frequency components combined into Mel-frequency, provides a concise representation of sensor data, which leads to robust recognition of human activities from wearable sensor readings.

The degree of variation within an activity signal distribution is characterized through its skewness with respect to its mean. When probability distribution of random variable is not symmetrical around its mean, it is considered to not be skewed

$$S_K = \frac{\sum_{m=0}^R \frac{(T_m - \bar{T})^3}{R}}{\sigma^3} \quad (3)$$

R represents the number of activity samples

$$S_s = \begin{cases} <0, & (-ve)ly \text{ skewed} \\ =0, & \text{symmetric value} \\ >0, & (+ve)ly \text{ skewed.} \end{cases} \quad (4)$$

Spectral Kurtosis is employed to evaluate the presence of outliers in the activity signal pattern. Kurtosis serves as a measure of a distribution’s deviation from the characteristics of a normal distribution, indicating its extremeness or flatness

$$S_K = \frac{\sum_{m=1}^R \frac{(T_m - \bar{T})^4}{R}}{\sigma^4} \quad (5)$$

T represents the mean value of activity signal, while σ represents the standard deviation of T . S_K is defined as the ratio between the fourth-order moment and the square of the second-order moment

$$S_K = \begin{cases} <0, & \text{Flat} \\ =0, & \text{Normal} \\ >0, & \text{Pleaked.} \end{cases} \quad (6)$$

The computation of the normalized pitch frequency (NPF) involves the utilization of the average autocorrelation function (AC) and average magnitude (AM) difference function. AC quantifies the peak magnitude within pitch period, while AM reveals the notch during that period. The NPF is a ratio of the pitch frequency to the maximum frequency present in the input activity signal

$$\text{NPF}(t) = \frac{\text{ac}(t)}{1 + \text{AM}(t)}. \quad (7)$$

Theoretically, these features (MFCCs, Mel-spectrogram, spectral skewness, spectral kurtosis, and NPF) are captured to

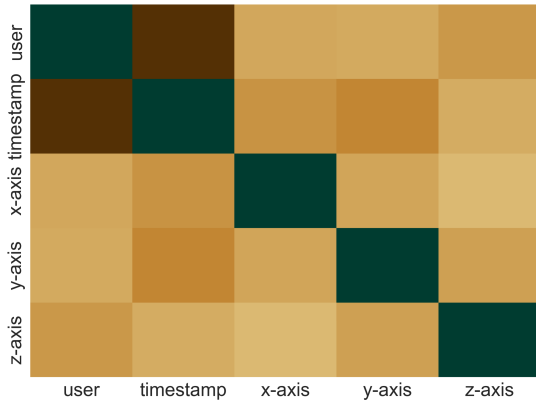


Fig. 3. Heatmap of WISDM dataset.

provide a comprehensive representation of human activities. This combination captured various aspects of the input sensor data, such as spectral characteristics, statistical properties, and frequency-related information. By using multimodal information fusion, the (Algorithm 1) enhanced the discriminative power of the model to distinguish between different activities more effectively. Moreover, the integration of a variety of features enhanced the robustness and generalization ability of the model.

Algorithm 1 Segmentation and Feature Extraction

```

1: Input: Sensor data  $D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)\}$ 
2: Output: Feature vectors  $F = \{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_M\}$ 
3:  $W \leftarrow$  Window size;  $R \leftarrow$  Overlap ratio
4:  $F \leftarrow$  Empty list for feature vectors
5: for  $i = 0$  to  $N - W$  with step  $\text{int}(W \times (1 - R))$  do
6:   segment  $\leftarrow \{(\mathbf{x}_{i,1}, \mathbf{x}_{i,2}, \dots, \mathbf{x}_{i,D}), y_i\}$ 
7:   sensor_readings  $\leftarrow \{(\mathbf{x}_{i,1}, \mathbf{x}_{i,2}, \dots, \mathbf{x}_{i,D-1})\}$ 
8:   mfccs  $\leftarrow$  ComputeMFCC(sensor_readings)
9:   kurtosis  $\leftarrow$  CalculateKurtosis(sensor_readings)
10:  skewness  $\leftarrow$  CalculateSkewness(sensor_readings)
11:  mel  $\leftarrow$  ComputeMelSpectrogram(sensor_readings)
12:  ac  $\leftarrow$  AutoCorrelation(sensor_readings)
13:  am  $\leftarrow$  AbsoluteDifference(Diff(sensor_readings))
14:  pitch_frequency  $\leftarrow$  ArgMax(ac)/Length(ac)
15:  max_frequency  $\leftarrow$  Max(am)
16:  npf  $\leftarrow$  pitch_frequency/max_frequency
17:  features  $\leftarrow \{\text{mfccs, kurtosis, skewness, mel, npf}\}$ 
18:  activity  $\leftarrow y_i$ 
19:   $\mathbf{f}_i \leftarrow \{\text{features, activity}\}$ 
20:   $F \leftarrow F \cup \{\mathbf{f}_i\}$ 
21: end for

```

D. Deep Capsule Network

The architecture of the DeepCapsNet employed for HAR comprises of following phases: convolutional phase, capsule network phase, and output layer as illustrated in Table V. Details of each phase are provided below.

1) *Convolutional Phase:* This phase consists of three 1-D CLs. A task was performed by every single layer like Conv operation, batch normalization (BN), and activation function. The overall structure of all Conv layers remains similar except

TABLE V

PARAMETERS OF THE DEEPCAPSNET FOR HAR

Layer Name	Kernel	Filters	Strides	Output Shape
Input	-	-	-	None $\times$ 144 $\times$ 1
Conv 1 & BN & ELU	5	64	1	None $\times$ 140 $\times$ 64
Conv 2 & BN & ELU	5	128	1	None $\times$ 136 $\times$ 128
Conv 3 & BN & ELU	3	256	1	None $\times$ 132 $\times$ 256
Dense Capsules	-	256	-	None $\times$ 2176 $\times$ 16
Lambda Capsule	-	-	-	None $\times$ 2176
Output	-	6	-	None $\times$ 6

for the kernel sizes—where the first two layers employed a kernel size of five, last layer used a size of three. The Conv operations extract the distinctive characteristics from the input. Thus, characteristics derived from Conv layers are concatenated to form input vectors of the dense capsule layer. The multiple Conv layers with different filter (kernel) sizes serve two purposes. First, these layers can capture numerous temporal dependencies, thereby enhancing the process of feature extraction. Second, the absence of pooling operations in these layers enables the extraction of higher level features. Therefore, the network without pooling operations ensures the preservation of integral features. Detailed information on Conv operation, BN, and activation function is presented below.

a) *Convolutional operation:* As the activity data are characterized by time-series signals, the 1-D CLs are utilized to capture deep distinctive features. Consider each activity signal is expressed as $X_n = [x_1, x_2, \dots, x_N] \in \mathbb{R}^{1 \times N}$, here N denotes the length of activity signal. The Conv operation in each CL layer involves convolving the input X (output of the previous layer H_j^{l-1}) to get the feature map $H_j^l \in \mathbb{R}^{K^l \times D^l}$, where D^l represents the dimensions, and K^l shows the number of feature maps of the l th CL. The result H_j^l of the j th feature map within the l th CL is computed as

$$H_j^l = \sum_{i=1}^{K^{l-1}} H_i^{l-1} w_{ji}^l + b_j^l \quad (8)$$

w_{ji}^l refers to the filter/kernel weight that connects the j th feature map in l th CL to the i th feature map in $(l-1)$ th CL, and b_j^l represents the associated bias.

b) *Batch normalization:* During network training, the constant updating of kernel weights leads to a dynamic adaptation of the input data distribution within each CL to capture the intricate patterns. So, updating the kernel weights reduces the pace of convergence during the training phase. In order to improve speed of training network and stability, a BN is used before employing activation function in every CL. Application of BN normalizes the output feature within every batch of every CL to stop the feature from reaching too large or too small. Thus, BN enhances the speech and accuracy of the network. BN is used as

$$\hat{H}_j^l = \gamma_c \frac{(H_j^l - E[H_j^l])}{\sqrt{\text{Var}[H_j^l + \epsilon]}} + \beta_c \quad (9)$$

where Var and E denote the variance and mean of H_j^l , respectively. γ_c and β_c are parameters that improve during the training of network. ϵ is a fixed value greater than zero.

c) *Activation function*: All three CLs utilizes an activation function to compute the output of feature map which is represented by $H_j^{\text{conv},l} = f(H_j^l)$. Here, f represents an activation function. This study employed exponential linear unit (ELU) function within each layer to compute the feature map. ELU function achieved higher recognition accuracy compared to other functions. Mathematically, the ELU is written as

$$f(\hat{H}_j^l) = \begin{cases} H_j^l, & \text{for } H_j^l \geq 0 \\ (e^{H_j^l} - 1), & \text{for } H_j^l < 0. \end{cases} \quad (10)$$

2) *Capsule Network Phase*: A capsule network is comprised of several capsules to capture both feature and relative positions of input. Unlike CNNs that operate on scalar values, capsule networks are vector-oriented that indicate the features and their spatial relationship. Thus, a collection of neurons is termed capsule and the connection between capsule layers is vector–vector. Each capsule’s output vector is derived through a sum of input vectors from the preceding layer, followed by processing through a nonlinear activation function. The capsule phase comprises a dense capsule (primary) layer and a lambda capsule layer. A detailed description of both layers is given below.

a) *Dense capsule layer*: This layer comprises multiple capsules and is derived from the outputs of the last conv operation. The outputs $H_j^{\text{conv},L}$ from the final CL are utilized as inputs for the dense capsule layer which transforms the feature maps into corresponding vectors. These vectors are lastly processed employing squashing function to confine their values within the range [0, 1]. The dense layer is linked to the lambda capsule layer through the utilization of transformation matrices.

b) *Lambda capsule layer*: Lambda capsule layer is employed to compute the L2 norm along the last axis that effectively normalizes the vector representations of each capsule. This normalization process enhances the stability of the network by constraining the magnitudes of capsule vectors. Through these operations, the Lambda capsule layer contributes to the network’s capacity to capture discriminative features and spatial relationships within the input data that facilitate robust HAR.

3) *Output Layer*: This layer accepts the results computed from Lambda capsule layer. The findings from output layer provides the probability values for each activity, where human activity with the highest probability is the recognition outcome of the network.

E. Evaluation Metrics

Precision, recall, accuracy, and F1- score were employed to assess the performance of DeepCapsNet for HAR by employing the confusion matrix (CM). The CM comprised of true-positive (TP), true-negative (TN) when model correctly classifies activities and false-positive (FP), and false-negative (FN) when model classifies incorrectly. Accuracy is determined by calculating the successfully classified activities from the total activity samples analyzed using the following equation:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \left(\frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \right)_i \quad (11)$$

where N represents the total samples of activity.

Precision is a commonly used metric to assess the efficacy of a method in addressing activity recognition challenges, especially in scenarios where the dataset contains a significant instance of false positives which can impact overall accuracy. It is computed by the proportion of accurately identified positive activity samples to the total of both accurately and inaccurately identified positive samples as given in the following equation:

$$\text{Precision} = \frac{1}{N} \sum_{i=1}^N \left(\frac{\text{TP}}{\text{TP} + \text{FP}} \right)_i \quad (12)$$

Recall is calculated by the proportion of accurately identified positive activity samples to the total of accurately identified positive samples and inaccurately identified negatives samples as given in the following equation:

$$\text{Recall} = \frac{1}{N} \sum_{i=1}^N \left(\frac{\text{TP}}{\text{TP} + \text{FN}} \right)_i \quad (13)$$

F1-score is commonly employed to mitigate the trade-offs between recall and accuracy. It is computed by taking weighted average of recall and precision as given in the following equation:

$$\text{F1-score} = \frac{1}{N} \sum_{i=1}^N 2 \times \left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right)_i \quad (14)$$

IV. EXPERIMENTS

The proposed system addresses the challenge of activity recognition by leveraging a combined feature set (MFCCs, Mel-spectrogram, S_S , S_K , and NPF) and a DeepCapsNet that integrates the strengths of CNNs and capsule layers. This approach is compared with the well-established LSTM and BiLSTM models. To determine the superiority of the proposed methodology for accurate real-time HAR, the performance of several models, including 1-D-CNN, LSTM, BiLSTM, and DeepConvLSTM is evaluated on three datasets. The percentage split technique was applied to evaluate the proposed DeepCapsNet for HAR. Specifically, 80% of the features were used for the training of DeepCapsNet and the 20% of features were used to test the network [37], [38]. The study [39] has reported that the optimal performance is achieved when 20%–30% of the features are utilized for testing the model, while the remaining features are used to train the model. This split ensures that the model achieves a valid and accurate performance.

A. DeepCapsNet Performance

In this study, the proposed capsule network enhanced the performance and minimized the loss for both training and validation activity samples, which shows the significance and effectiveness of the proposed network as demonstrated in Fig. 4.

This study employed three datasets including UCI HAR, WISDM, and PAMAP2 to evaluate the performance of DeepCapsNet. The recognition performance of proposed network using UCI HAR, WISDM, and PAMAP2 datasets is presented in Tables VI–VIII, respectively.

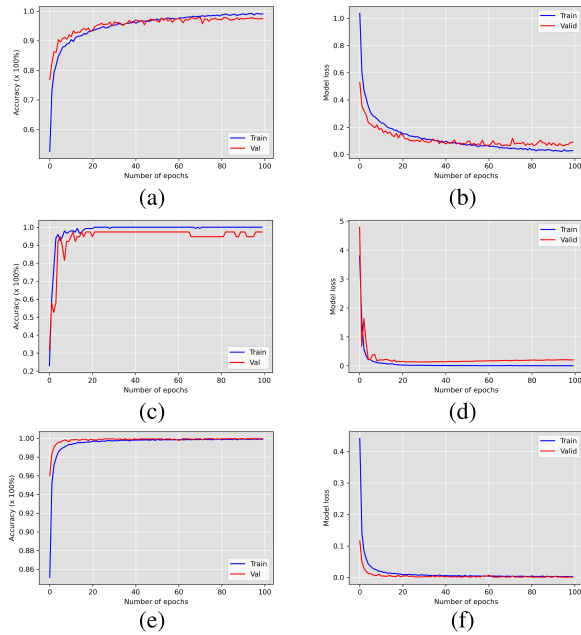


Fig. 4. Performance of DeepCapsNet. (a) Validation and Training accuracy on UCI HAR dataset. (b) Validation and Training loss on UCI HAR dataset. (c) Validation and Training accuracy on WISDM dataset. (d) Validation and Training loss on WISDM dataset. (e) Validation and Training accuracy on PAMAP2 dataset. (f) Validation and Training loss on PAMAP2 dataset.

TABLE VI

PERFORMANCE (%) OF PROPOSED NETWORK FOR UCI HAR

Activity	Performance Metric			
	Accuracy	Precision	Recall	F1-score
DOWNSTAIRS	100	100	100	100
LAYING	100	100	100	100
SITTING	100	88.9	100	94.0
STANDING	85.7	100	85.7	92.0
UPSTAIRS	100	100	100	100
WALKING	100	100	100	100
AVERAGE	97.6	98.2	97.6	97.4

TABLE VII

PERFORMANCE (%) OF PROPOSED NETWORK FOR WISDM

Activity	Performance Metric			
	Accuracy	Precision	Recall	F1-score
DOWNSTAIRS	97.8	97.1	97.8	97.0
JOGGING	96.9	97.6	96.9	97.0
SITTING	100	100	100	100
STANDING	100	97.2	100	99.0
UPSTAIRS	98.0	99.5	98.0	99.0
WALKING	98.1	100	98.1	99.0
AVERAGE	98.5	98.6	98.5	98.7

The table shows the precision, recall, F1-measure, and recognition accuracy of all activities in three datasets and reveals the reliability of DeepCapsNet compared to the baseline models [35], [40], [41], [42]. The proposed DeepCapsNet for UCI HAR dataset achieved 97.6% recognition and outperformed the existing model [35] for UCI HAR for all human activities in UCI HAR. The highest F1-score of 100%,

TABLE VIII

PERFORMANCE (%) OF PROPOSED NETWORK FOR PAMP2

Activity	Performance Metric			
	Accuracy	Precision	Recall	F1-score
CYCLING	100	100	100	100
DOWNSTAIRS	100	100	100	100
IRONING	100	100	100	100
JUMPING	100	99.9	100	100
LYING	99.9	99.9	99.9	100
NORDIC WALK	100	100	100	100
RUNNING	99.9	100	99.9	100
SITTING	100	100	100	100
STANDING	100	100	100	100
UPSTAIRS	100	100	100	100
VACUUM	100	100	100	100
WALKING	100	100	100	100
AVERAGE	99.9	100	100	100

100%, and 100% were achieved for downstairs, lying, and walking, respectively, and standing achieved the lowest F1-score of 92%. Likewise, highest precision values of 100%, 100%, and 100% were for downstairs, lying, and standing, respectively, while the low precision value of 88.9% was obtained for sitting. The interactivity CM of DeepCapsNet using UCI HAR dataset is shown in Fig. 5(a). The figure shows the correctly recognized samples diagonally and confusion among activities in respective rows. The Fig. 5(b) of proposed network for WISDM dataset demonstrates that the highest recognition accuracy achieved for sitting, standing, and waking were 100%, 100%, and 98.1%, respectively. The upstairs and downstairs were also classified with enhanced performance compared to the baseline models for WISDM [41] and weighted accuracy of DeepCapsNet also beat the existing techniques [40], [41], [42], [43]. In the Fig. 5(c) for PAMAP2 dataset achieved the 100%, 100%, 100%, and 100% accuracy for cycling, downstairs, ironing, and jumping, respectively. The minimum accuracy of 99% was obtained for lying. HAR models proposed in the existing studies for UCI HAR obtained low recognition accuracy for the “vacuum” activity because of the mix-up characteristics with “vacuum” activity, nonetheless the proposed network classified each activity from related features with improved performance. The proposed DeepCapsNet predicted each activity in all datasets with an improved rate due to the relevant feature and DeepCapsNet architecture.

B. Comparisons With DL Models

To construct the HAR model, the extracted features were given to different DL models: CNN, LSTM, BiLSTM, and DeepConvLSTM. Moreover, 12 experiments (three datasets $\times$ four DL models) were conducted to evaluate the performance of DL models coupled with extracted features. Table IX exhibits how the proposed DeepCapsNet outperformed other DL models, with average accuracy for UCI HAR, WISDM, and PAMAP2 of 97.6%, 98.5%, and 100%, respectively. The DeepConvLSTM and BiLSTM models achieved an accuracy of 97.5% and 91.3%, respectively, using the PAMAP2 dataset, as opposed to 92.7% for CNN, and 88.5%

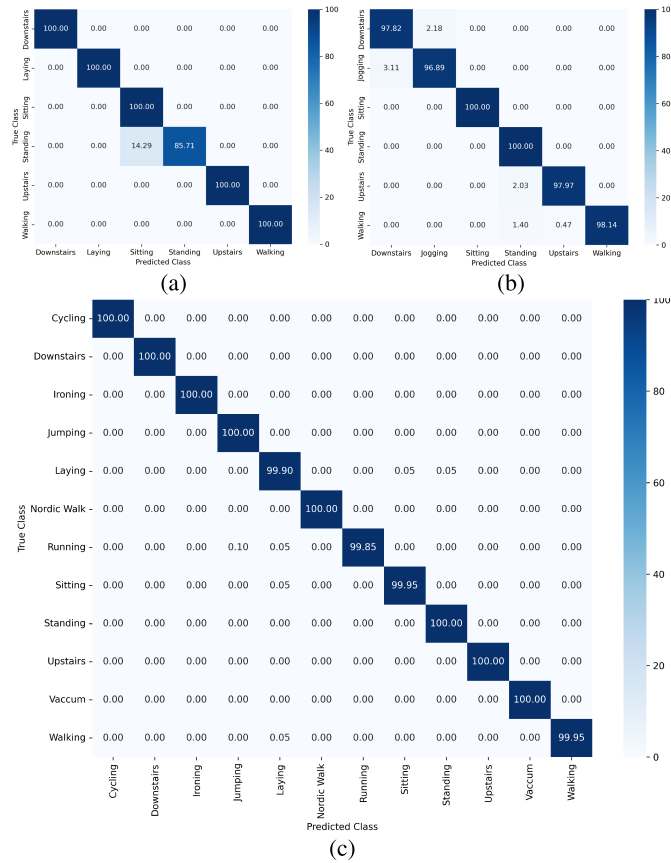


Fig. 5. Confusion matrices for (a) UCI HAR, (b) WISDM, and (c) PAMAP2. The y-axis represent the true class, while the x-axis displays the predicted class.

TABLE IX

COMPARATIVE ANALYSIS OF PROPOSED DEEPCAPSNET WITH DL MODELS

DL Model	Accuracy (%) for each dataset		
	UCI HAR	WISDM	PAMAP2
CNN	90.1	89.4	92.7
LSTM	73.5	86.8	88.5
BiLSTM	75.6	87.3	91.3
DeepConvLSTM	93.2	86.8	97.5
Proposed Model	97.6	98.5	100

for LSTM. Furthermore, utilizing WISDM and UCI HAR datasets, DeepConvLSTM model outperformed the CNN, LSTM, and BiLSTM models. In conclusion, the proposed DeepCapsNet for HAR beat the other four DL models using all three datasets and obtained the highest average accuracy.

C. Comparisons Analysis With Baseline Studies

To exhibit the significance and reliability of the proposed DeepCapsNet for HAR, this study compared the achieved performance on aforementioned datasets with the baseline studies. Table X provides a comprehensive summary of the comparison analysis. The recognition accuracy is noticeably better than baseline studies, which show the significance of proposed DeepCapsNet for HAR. However, in some cases, the recognition accuracy of the DeepCapsNet for particular human activity is lower compared to the existing studies. For instance, the proposed model in [35] achieved 94% accuracy for the

TABLE X

COMPARISON OF DEEPCAPSNET HAR AND BASELINE HAR MODELS USING THE UCI HAR, WISDM, AND PAMAP2 DATASETS

Study	Dataset	Accuracy (%) of each Activity													Avg
		HA ₁	HA ₂	HA ₃	HA ₄	HA ₅	HA ₆	HA ₇	HA ₈	HA ₉	HA ₁₀	HA ₁₁	HA ₁₂	HA ₁₃	
[40]	WISDM	72	87	83	93	98	98	x	x	x	x	x	x	x	93
	UCI HAR	99	99	94	95	x	99	x	x	x	x	x	x	x	98
[35]	PAMAP2	90	94	90	91	x	91	84	96	89	95	96	86	89	91
	UCI HAR	91	86	88	94	x	98	x	98	x	x	x	x	x	93
[41]	WISDM	87	89	96	98	98	98	x	x	x	x	x	x	x	96
	UCI HAR	96	96	90	98	x	98	x	100	x	x	x	x	x	96
[42]	PAMAP2	82	94	100	97	x	98	93	100	99	100	82	87	98	94
	WISDM	88	91	98	100	98	99	x	x	x	x	x	x	x	97
[43]	PAMAP2	98	93	90	99	x	99	98	99	94	100	100	97	96	97
	WISDM	99	98	95	99	99	100	x	x	x	x	x	x	x	97
this	UCI HAR	98	95	98	93	x	99	x	100	x	x	x	x	x	97
	PAMAP2	90	96	97	100	x	100	97	98	95	75	97	100	95	95
this	WISDM	98	98	100	100	97	98	x	x	x	x	x	x	x	98.5
	UCI HAR	100	100	100	86	x	100	x	100	x	x	x	x	x	98
this	PAMAP2	100	100	100	100	x	100	100	100	100	100	94	100	100	100

HA₁=Upstairs, HA₂=Downstairs, HA₃=Sitting, HA₄=Standing, HA₅=Jogging, HA₆=Walking, HA₇=Vacuum, HA₈=Lying, HA₉=Running, HA₁₀=Nordic Walk, HA₁₁=Jumping, HA₁₂=Ironing, HA₁₃=Cycling

TABLE XI

COMPARISON OF TRAINING TIME (SECONDS) WITH BASELINE MODELS

Model	UCI HAR	WISDM
DCapsNet [44]	1,226	1,971
CNN [45]	855	1,709
CNN+LSTM [45]	17,499	31,500
CNN-DTCWT [46]	1,006	1,907
Proposed Model	455.4	682.2

standing activity in UCI HAR dataset, while the proposed capsule network achieved 86% accuracy for the standing activity. Nevertheless, the DeepCapsNet outperformed baseline studies by achieving 98% average accuracy and classified each human activity with high accuracy, low computational time. This concludes that the DeepCapsNet for HAR is more reliable and accurate compared to the baseline studies.

D. Computational Time

The training process is utilized to measure the complexity and computational efficiency of the DeepCapsNet. The training time is directly proportional to the amount of activity data. If data are in large amount, the training time of model costs a lot. The computational efficiency using the training time is compared with mentioned models in Table XI. It can be observed that the proposed model consumed less training time. The complexity of the given hybrid models [refs] increased and therefore the computational time also increased. The proposed model achieved promising results in terms of both accuracy and computational efficiency.

V. CONCLUSION

The challenges in HAR lie in capturing robust and discriminative features for accurate recognition of person physical activity. In this paper, a distinct features set are extracted to capture a broad array of information along with patterns from sensors data. Then the derived features were employed as input to DeepCapsNet. The DeepCapsNet consists of CLs, with each CL follows a BN layer in order to accelerate convergence. After these layers, the primary capsule (dense) layer in

integrated with the lambda capsule layer. The CLs are utilized to capture temporal representation along with scalar features of sensors data which are given to capsule layers to obtain probability of each activity. The DeepCapsNet is evaluated on three publicly available datasets: UCI HAR, WISDM, and PAMAP2, achieving recognition accuracy of 97.6%, 98.5%, and 100%, respectively. The experimental results have revealed DeepCapsNet as proposed by the researcher achieves a more better accuracy as compared to other DL models of HAR such as 1-D-CNN, LSTM, BiLSTM, and DeepConvLSTM. The performance comparison with baseline HAR models demonstrates the robustness and significance of the DeepCapsNet model. The achieved recognition accuracy also demonstrates that DeepCapsNet is well suited for all three public datasets. The limitation of proposed approach lies in its dependency on domain experts to capture and merge diverse features. Although this approach demonstrates excellent performance across all three datasets, its applicability to other domains remains unexplored. In future, the proposed model can be applied to identify irregular patterns such as temperature fluctuations or pollution spikes using environmental sensors data. Second, the proposed model can enable predictive models for forecasting weather changes and support real-time environmental monitoring for proactive interventions. The model offers a robust framework to address environmental challenges and improve decision-making in environmental management using real-time signal processing. Finally, user experience studies will be conducted to evaluate the usability of the model in real-world application environments.

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